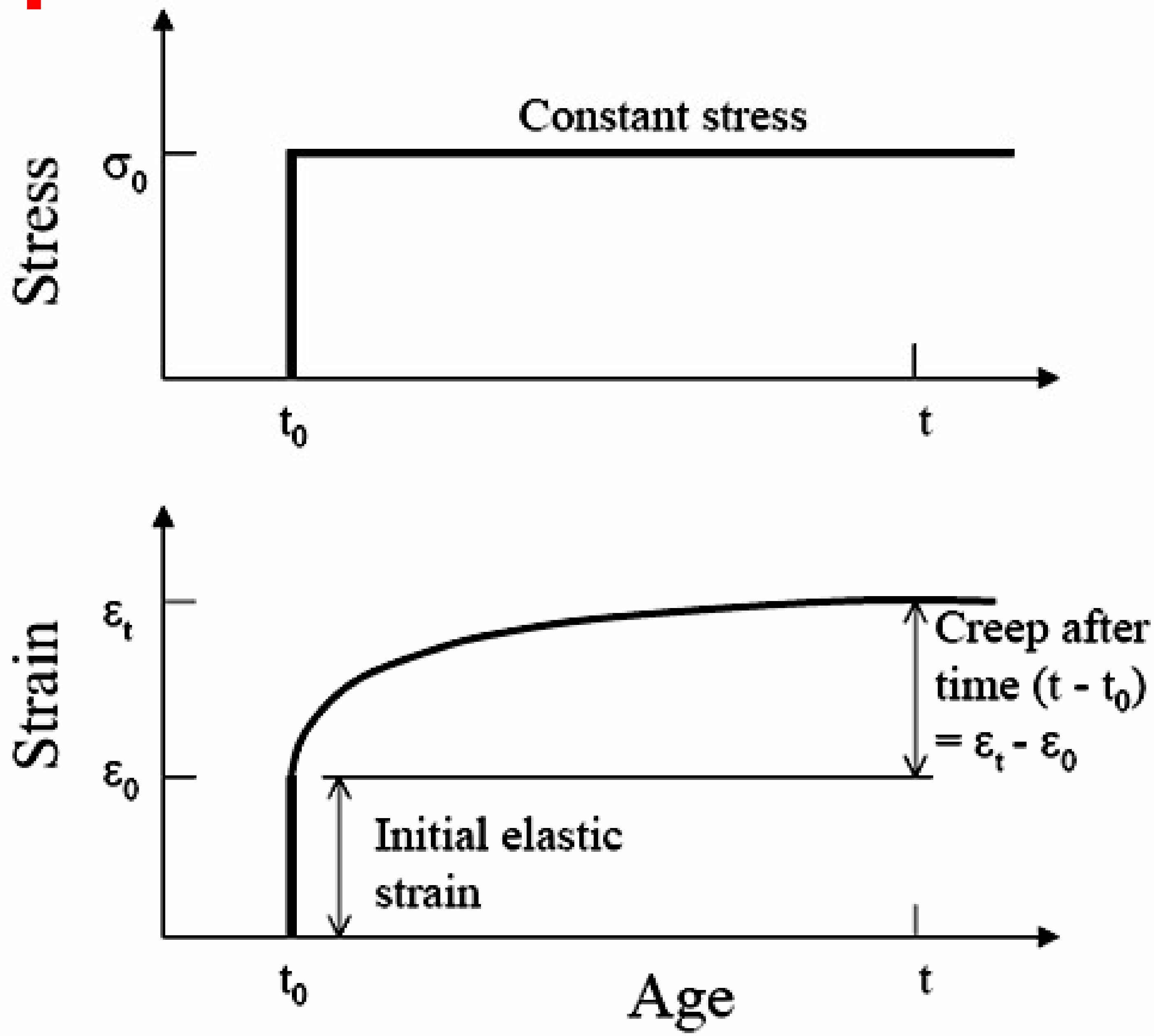
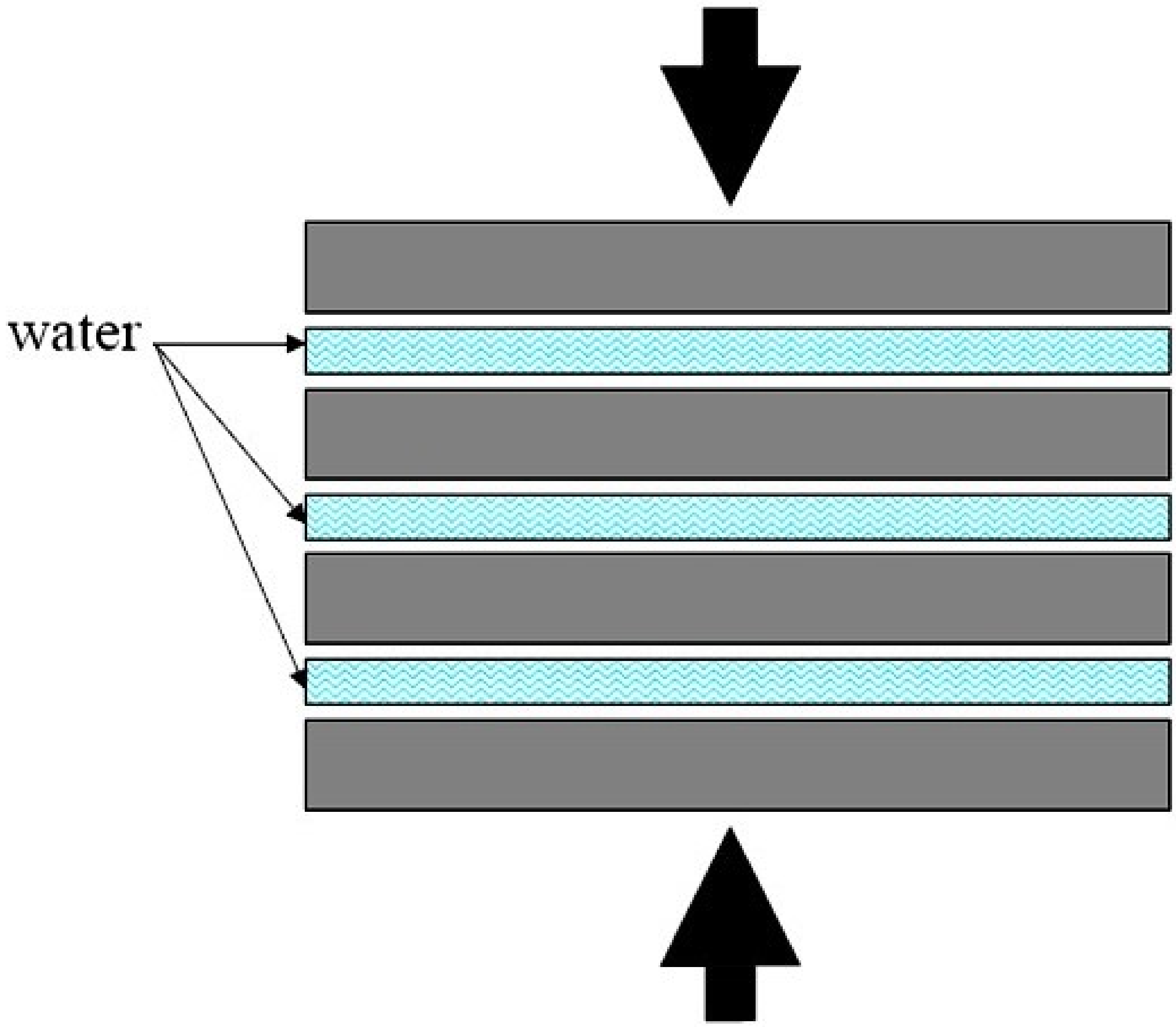
Creep

- **Creep** is defined as the increase of strain in concrete with time under sustained stress

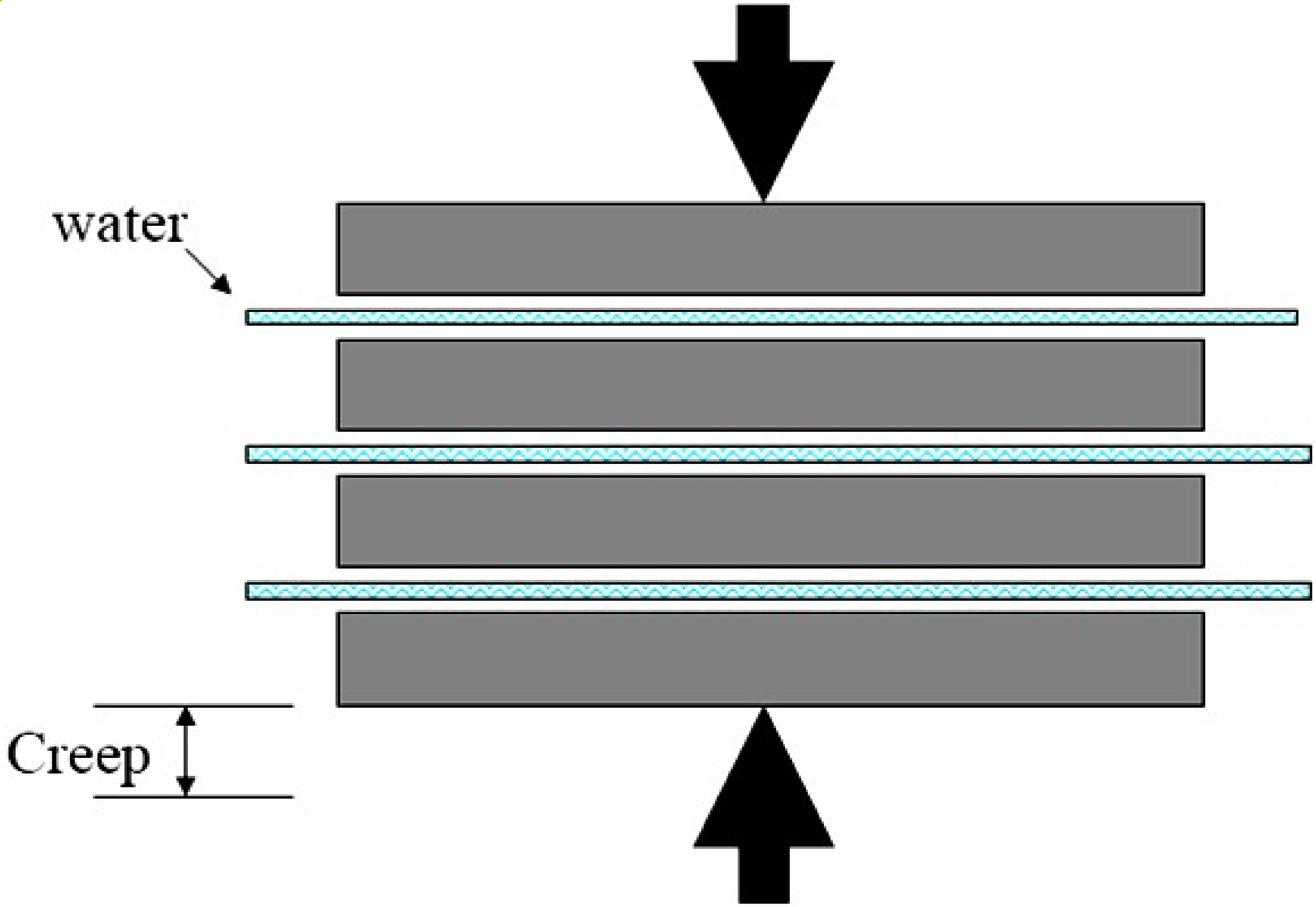
Creep Under a Constant Stress



Creep in Concrete



Creep in Concrete

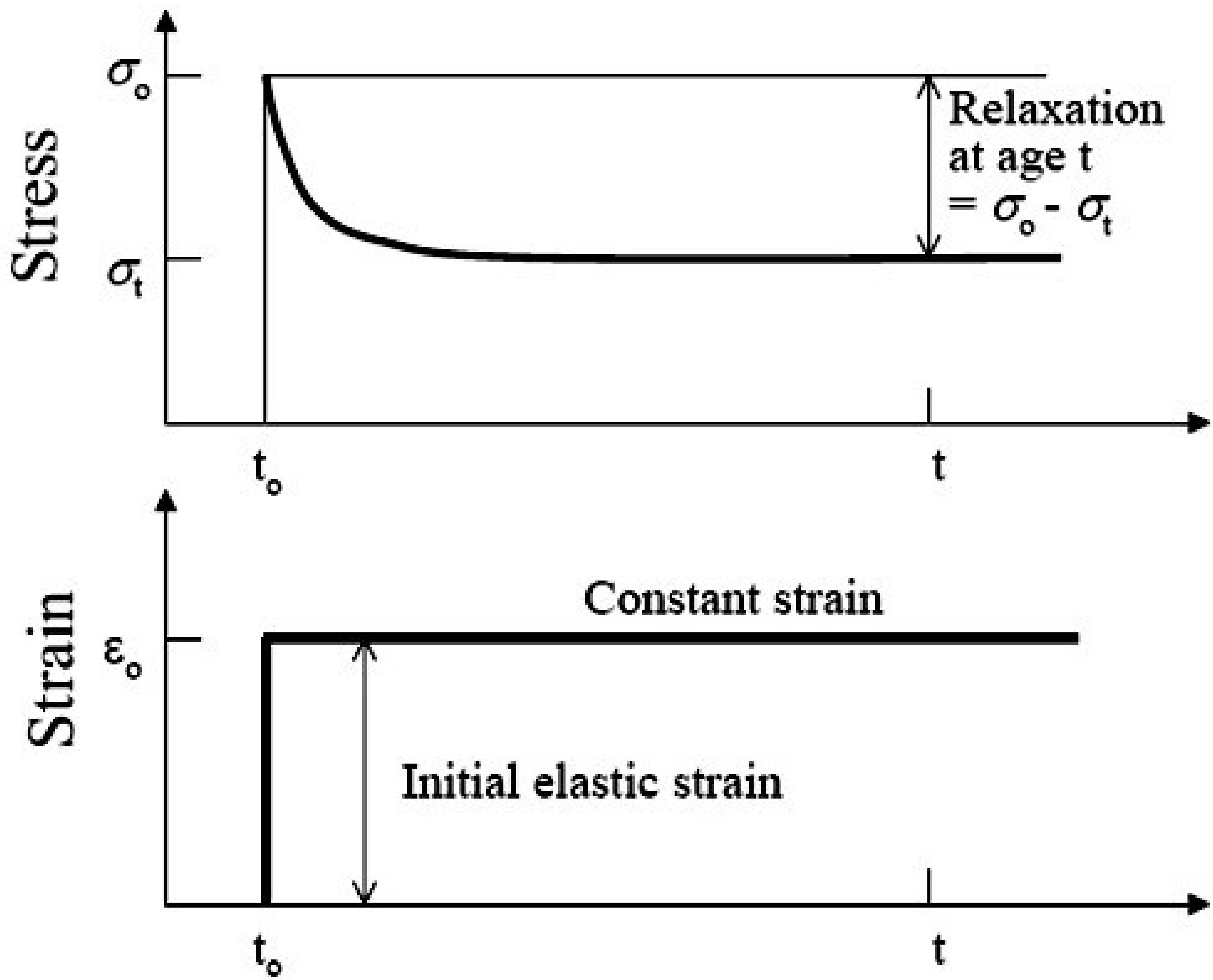


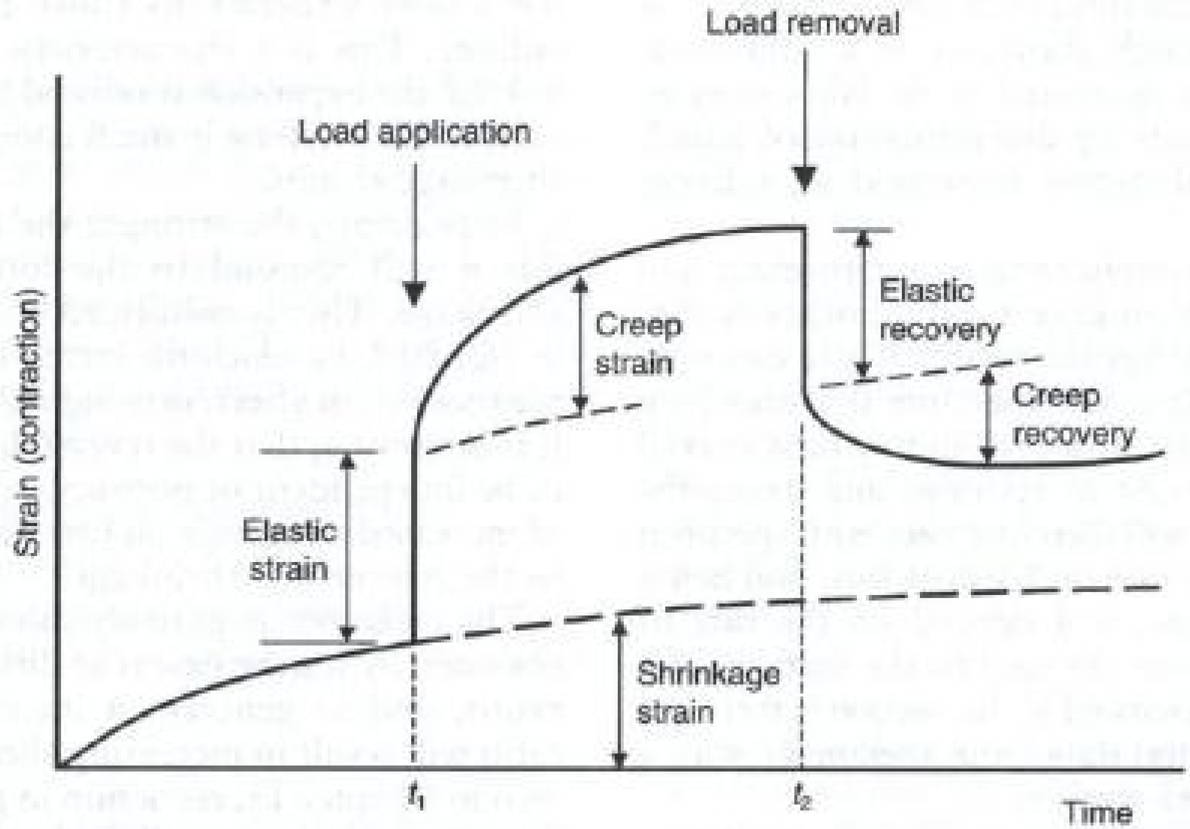
Creep

- If a loaded specimen is being subjected to a constant strain, the creep decreases the stress progressively with time. This is called **relaxation**

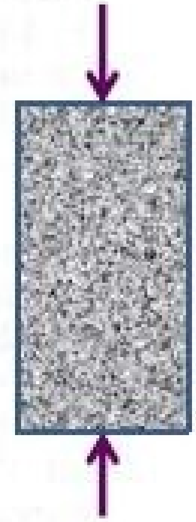
- All the factors which influence shrinkage influence creep also in similar way

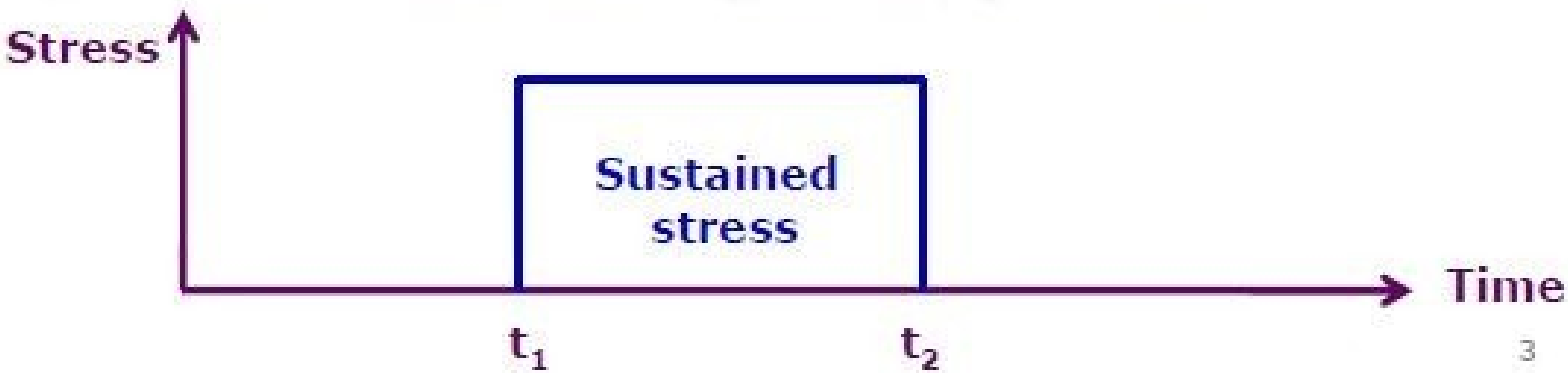
Relaxation for Concrete





The response of concrete to a compressive stress applied in a drying environment.





Effects of Creep

- It affects strain, deflection and stress distribution in reinforced concrete structures
- For example: Creep of concrete increases the deflection of reinforced concrete beams and, in some cases, may be a critical consideration in design

Factors Affecting Shrinkage and Creep

- Shrinkage/Creep increases with
  - Decreasing aggregate volume fraction
  - Decreasing modulus of elasticity of aggregate
  - Increasing water-to-cement ratio
  - Decreasing relative humidity